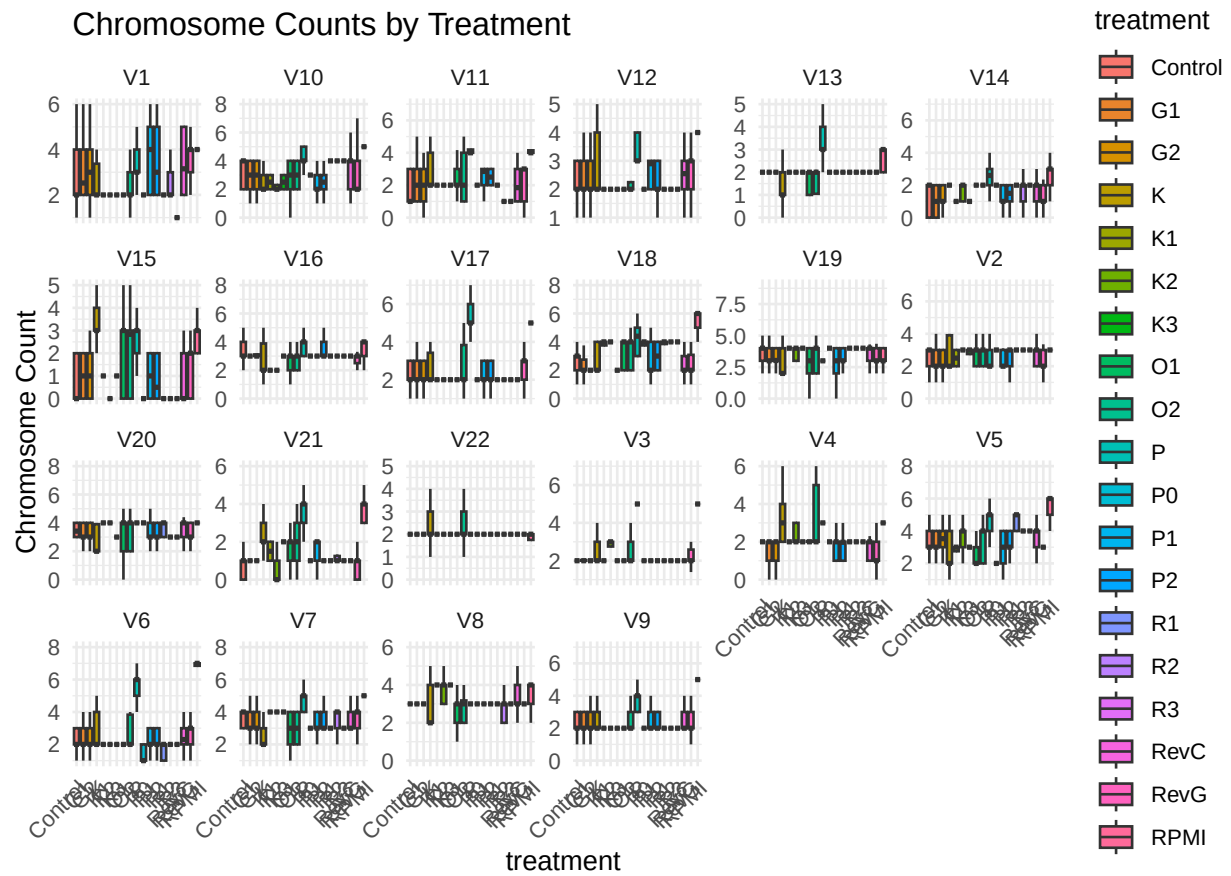
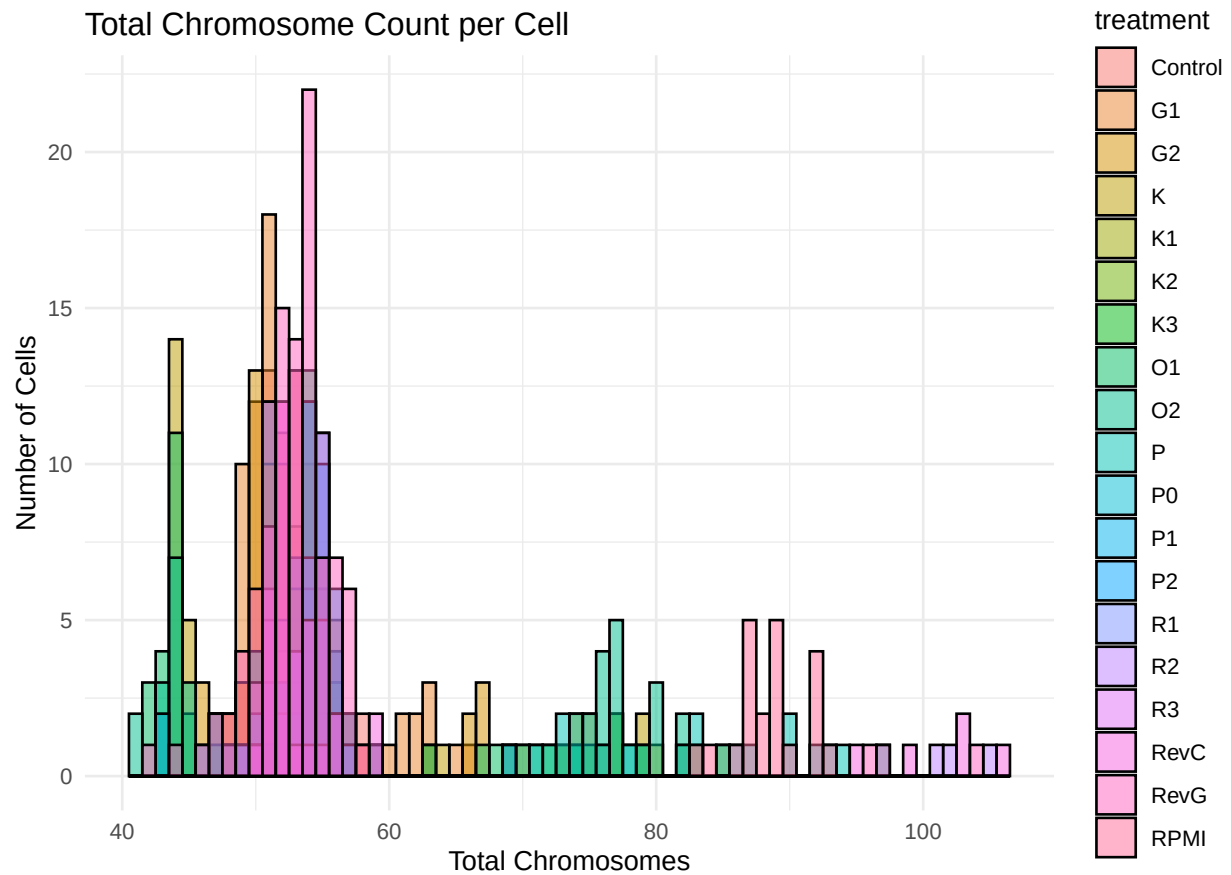


Permanova

Olofin Daniel

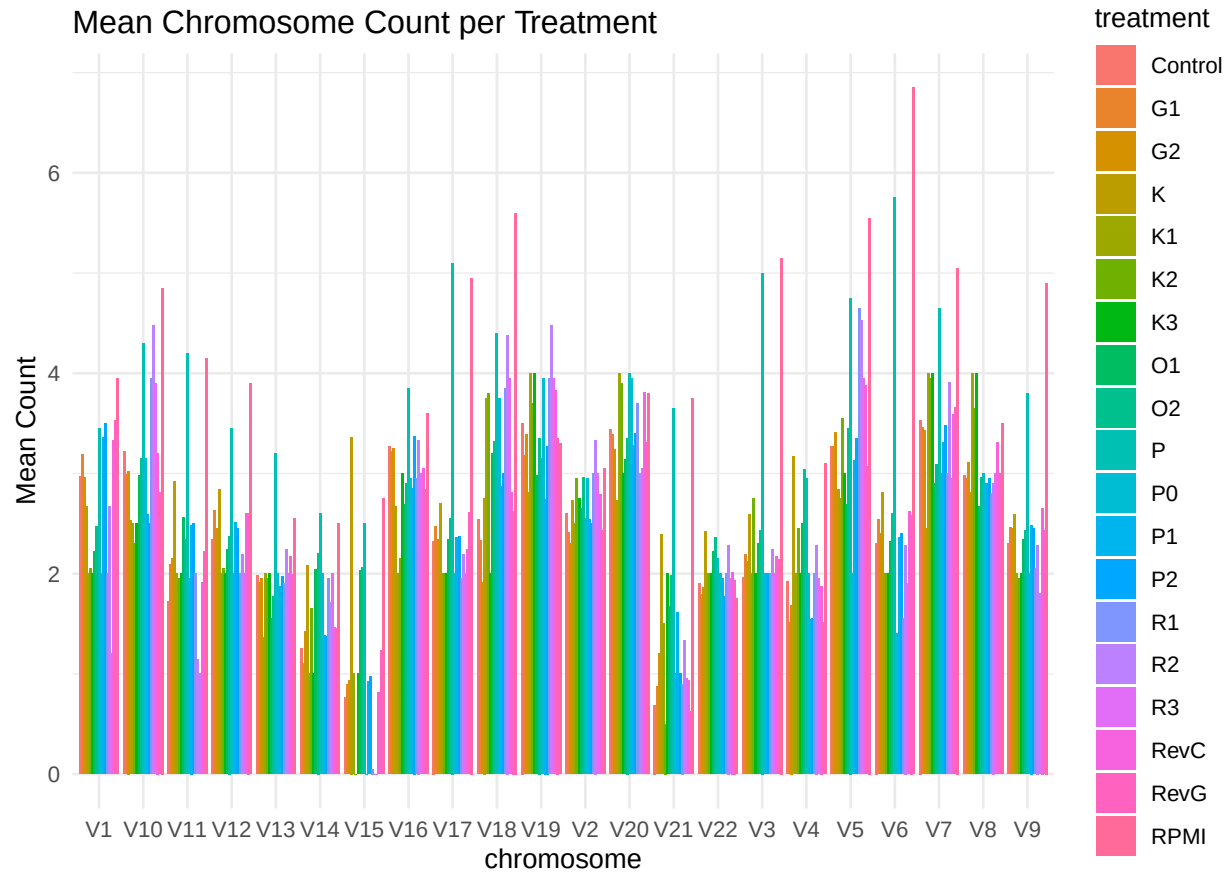
2025-06-06





```
# Mean per chromosome per treatment
mean_chr <- karyo_long %>%
  group_by(treatment, chromosome) %>%
  summarise(mean_count = mean(count, na.rm = TRUE), .groups = "drop")

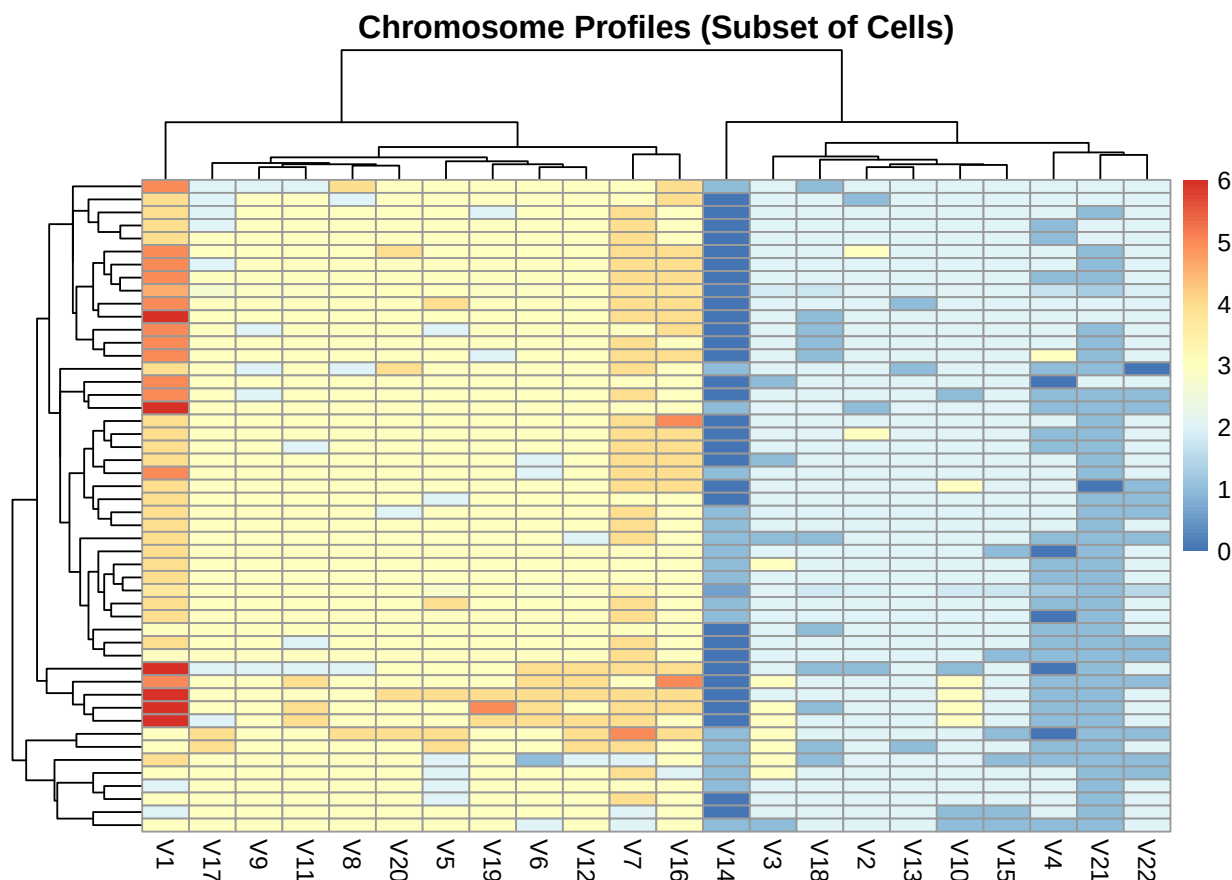
ggplot(mean_chr, aes(x = chromosome, y = mean_count, fill = treatment)) +
  geom_bar(stat = "identity", position = "dodge") +
  theme_minimal() +
  labs(title = "Mean Chromosome Count per Treatment",
       y = "Mean Count")
```



```
library(pheatmap)

# Make a cell-by-chromosome matrix
karyo_mat <- as.matrix(karyo_df[, paste0("V", 1:22)])
rownames(karyo_mat) <- karyo_df$cloneID

# Simple heatmap
pheatmap(karyo_mat[1:50, ], cluster_rows = TRUE, cluster_cols = TRUE,
         main = "Chromosome Profiles (Subset of Cells)",
         show_rownames = FALSE)
```



Step 5: Run PERMANOVA - testing treatment effect after accounting for cell line differences

```
adonis2(dist_flask ~ line + treatment,
  data = meta_flask,
  strata = meta_flask$line,      # Keep permutations within each line
  permutations = 999)
```

```
## Permutation test for adonis under reduced model
```

```
## Blocks: strata
```

```
## Permutation: free
```

```
## Number of permutations: 999
```

```
##
```

```
## adonis2(formula = dist_flask ~ line + treatment, data = meta_flask, permutations = 999, strata = meta_flask$line)
```

```
##          Df SumOfSqs      R2      F Pr(>F)
```

```
## Model     20   5943.7 0.76725 3.2964  0.617
```

```
## Residual  20   1803.1 0.23275
```

```
## Total     40   7746.8 1.00000
```

Interpretation: In this analysis, we tested whether different treatment conditions led to statistically significant changes in karyotype composition across independently cultured flasks, while controlling for variation due to cell lines. The PERMANOVA result ($F = 3.296$, $p = 0.638$, $R^2 = 0.767$) indicates that although the model explains a large proportion of the variance in karyotype profiles (77%), this variation is primarily attributed to inherent differences between cell lines rather than treatment effects. The high p-value (> 0.05) suggests there is no statistically significant treatment effect on global karyotype composition at the flask level. Thus, under the current design and data structure, the treatments applied did not induce repro-

ducible, detectable changes in the overall chromosome profiles when adjusting for baseline variability across cell lines.

- 2: GLMM for Per-Chromosome Changes

```
library(lme4)

# Binary outcome: gain defined as >2 copies
karyo_df$gain_chr08 <- ifelse(karyo_df$V8 > 2, 1, 0)

# GLMM: treatment as fixed effect, replicate as random effect
model_chr8 <- glmer(gain_chr08 ~ treatment + (1 | replicate_id),
                    data = karyo_df, family = binomial)

summary(model_chr8)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: gain_chr08 ~ treatment + (1 | replicate_id)
## Data: karyo_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    462.2    555.2   -211.1    422.2      755
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.0700  0.1072  0.2072  0.3074  0.7556
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## replicate_id (Intercept) 3.525    1.877
## Number of obs: 775, groups: replicate_id, 41
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   4.000e+00  1.405e+00  2.848  0.00441 **
## treatmentG1   -7.092e-01  1.800e+00 -0.394  0.69359
## treatmentG2   -6.777e-01  1.780e+00 -0.381  0.70341
## treatmentK    -4.121e+00  2.084e+00 -1.977  0.04801 *
## treatmentK1    1.917e+01  4.794e+04  0.000  0.99968
## treatmentK2    2.300e+01  1.588e+05  0.000  0.99988
## treatmentK3    1.837e+01  3.220e+04  0.001  0.99954
## treatmentO1   -2.995e+00  1.874e+00 -1.598  0.10994
## treatmentO2   -2.234e+00  1.914e+00 -1.167  0.24323
## treatmentP    -2.780e+00  2.400e+00 -1.158  0.24682
## treatmentP0   -1.545e+00  2.462e+00 -0.627  0.53035
## treatmentP1   -1.661e+00  2.010e+00 -0.826  0.40863
## treatmentP2   -6.055e-01  2.075e+00 -0.292  0.77050
## treatmentR1   -2.470e+00  2.410e+00 -1.025  0.30531
## treatmentR2   -3.424e+00  2.386e+00 -1.435  0.15137
## treatmentR3    1.775e+01  1.154e+04  0.002  0.99877
## treatmentRevC -3.730e-01  1.924e+00 -0.194  0.84625
```

```
## treatmentRevG -1.385e+00  1.652e+00 -0.838  0.40186
## treatmentRPMI -1.569e+00  2.462e+00 -0.637  0.52388
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## optimizer (Nelder_Mead) convergence code: 0 (OK)
## unable to evaluate scaled gradient
## Model failed to converge: degenerate Hessian with 2 negative eigenvalues
```

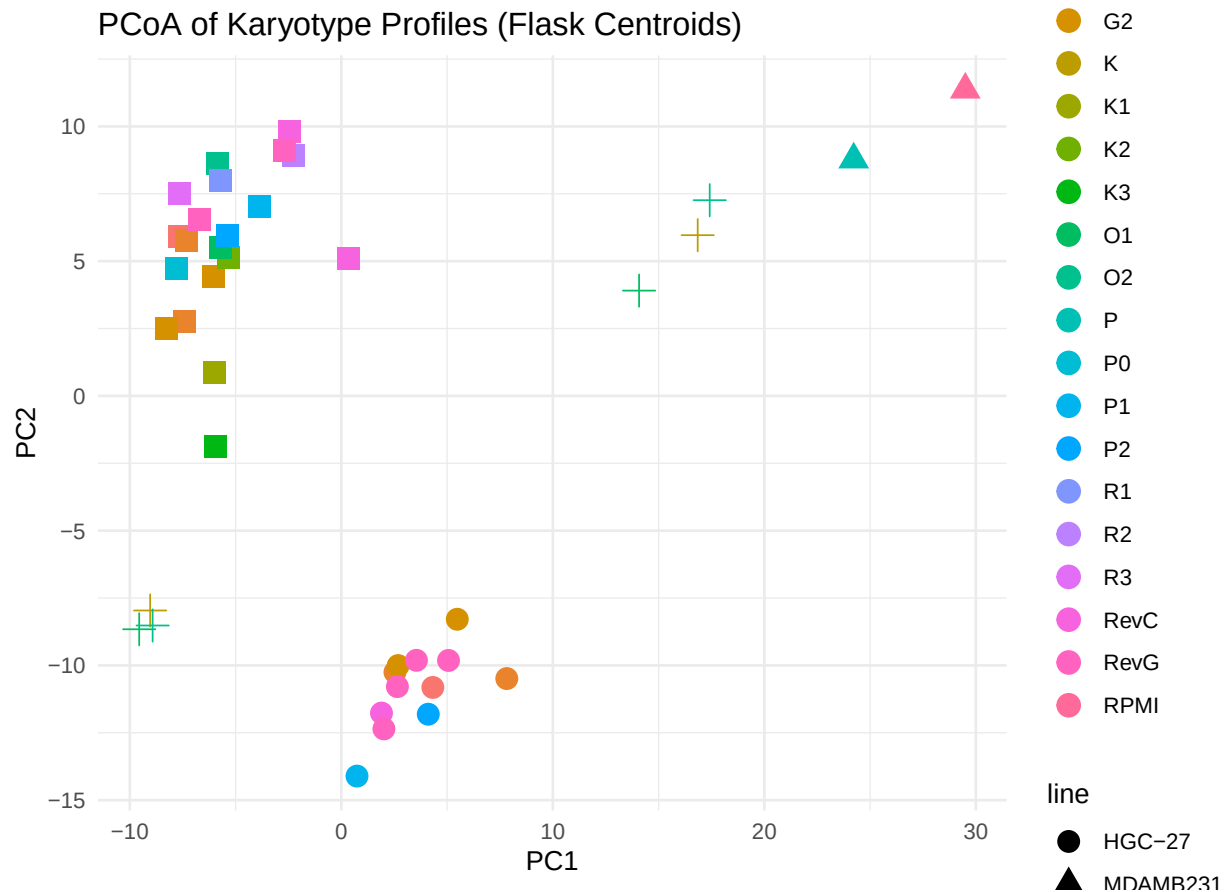
```
# Step 6: Test if treatment affects karyotype variability (dispersion)
disp_test <- betadisper(dist_flask, group = meta_flask$treatment, bias.adjust = TRUE)
anova(disp_test)
```

```
## Analysis of Variance Table
##
## Response: Distances
##           Df Sum Sq Mean Sq F value Pr(>F)
## Groups      9  492.46   54.718   0.9379 0.5132
## Residuals  22 1283.52   58.342
```

Interpretation: We assessed the assumption of homogeneity of multivariate dispersions using the `betadis-`
`per()` function. The analysis revealed no significant difference in the variability of karyotype composition
across treatment groups ($F(9, 22) = 0.94$, $p = 0.513$), indicating that group dispersions are comparable.
This supports the validity of our PERMANOVA results, as the observed group differences are unlikely to be
driven by unequal dispersion.

```
# Step 7: Visualize results using PCoA (ordination plot)
pcoa <- cmdscale(dist_flask, eig = TRUE, k = 2)
pcoa_df <- data.frame(PC1 = pcoa$points[,1],
                     PC2 = pcoa$points[,2],
                     treatment = meta_flask$treatment,
                     line = meta_flask$line)

ggplot(pcoa_df, aes(x = PC1, y = PC2, color = treatment, shape = line)) +
  geom_point(size = 4) +
  theme_minimal() +
  ggtitle("PCoA of Karyotype Profiles (Flask Centroids)")
```



```
# Load libraries
library(ggplot2)
library(vegan)
library(dplyr)

# Assume these data frames are already created:
# - flask_centroids: the averaged chromosome profile for each replicate_id
# - meta_flask: metadata with columns replicate_id, line, treatment

# Merge centroids and metadata
flask_data <- cbind(meta_flask, flask_centroids)

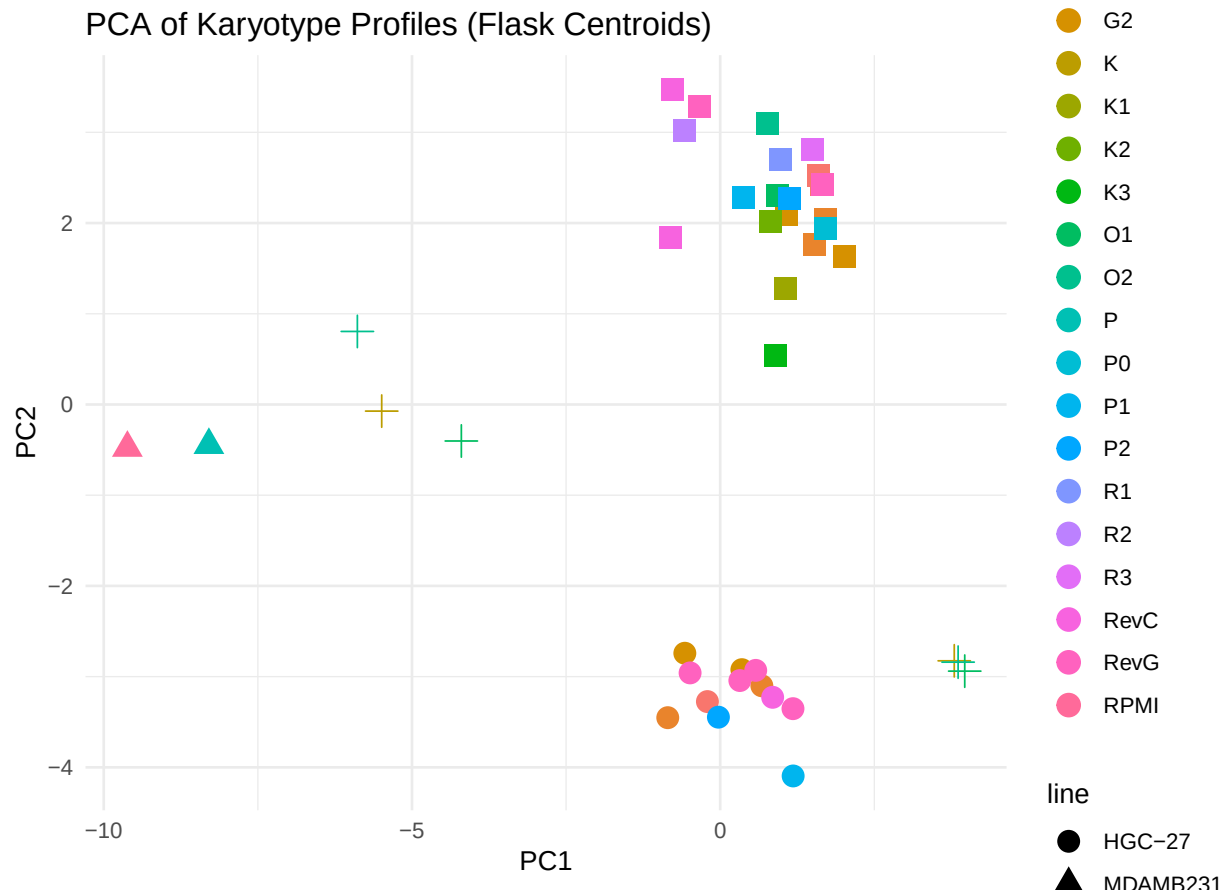
# Scale chromosome features
scaled_centroids <- scale(flask_centroids)

# PCA
pca_res <- prcomp(scaled_centroids)

# Add PCA scores to data frame
pca_df <- as.data.frame(pca_res$x[, 1:2])
pca_df <- cbind(meta_flask, pca_df)

# Plot PCA
ggplot(pca_df, aes(x = PC1, y = PC2, color = treatment, shape = line)) +
  geom_point(size = 4) +
```

```
labs(title = "PCA of Karyotype Profiles (Flask Centroids)",
     x = "PC1", y = "PC2") +
theme_minimal()
```



```
# MDS using Manhattan Distance
```

```
# Compute Manhattan distance
```

```
dist_flask <- vegdist(flask_centroids, method = "manhattan")
```

```
# Classical MDS (PCoA)
```

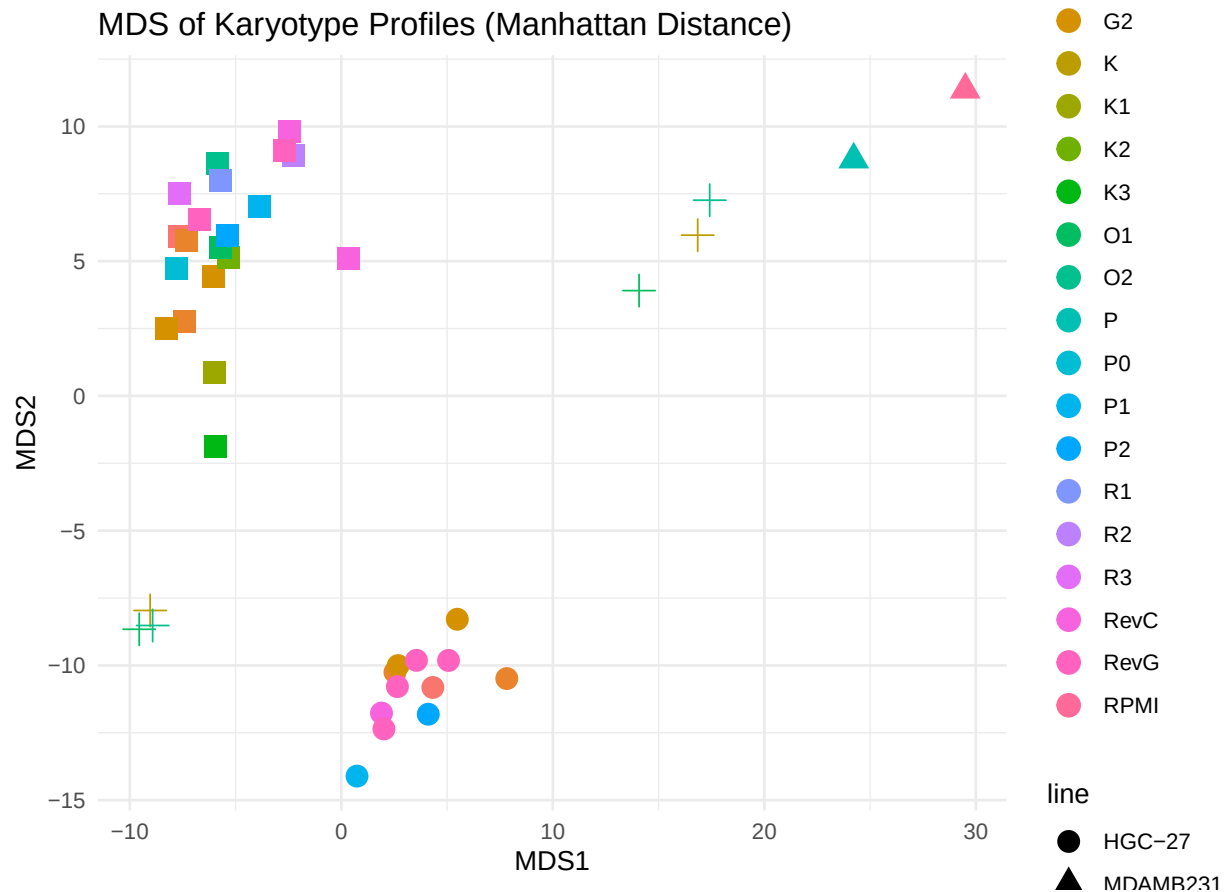
```
mds <- cmdscale(dist_flask, k = 2, eig = TRUE)
```

```
mds_df <- data.frame(MDS1 = mds$points[,1], MDS2 = mds$points[,2])
```

```
mds_df <- cbind(meta_flask, mds_df)
```

```
# Plot MDS
```

```
ggplot(mds_df, aes(x = MDS1, y = MDS2, color = treatment, shape = line)) +
  geom_point(size = 4) +
  labs(title = "MDS of Karyotype Profiles (Manhattan Distance)",
       x = "MDS1", y = "MDS2") +
  theme_minimal()
```

Findings:

Following best practices for single-cell karyotype analysis, we aggregated profiles at the replicate (flask) level and performed PERMANOVA to assess whether treatment influences global karyotype composition, adjusting for baseline differences across cell lines and restricting permutations within lines. To validate the test's assumptions, we assessed the homogeneity of dispersion using `betadisper()`. The result ($F = 0.94$, $p = 0.513$) indicated no significant difference in within-group variability across treatments, confirming that PERMANOVA assumptions were met. Thus, any detected differences—or lack thereof—in karyotype composition across treatments can be confidently interpreted as genuine shifts in group centroids rather than variability artifacts.

Next Step:

1. Fit GLMMs on individual chromosome gains/losses

- **Reason:** PERMANOVA looks at overall karyotype composition, but it can miss changes that occur only in specific chromosomes. So we fit Generalized Linear Mixed Models (GLMMs) to ask: **Does treatment X increase the odds of gaining chromosome 8, for example?**

```
library(lme4)
library(dplyr)
```

```

# Example for Chromosome 8 gain (assuming you have a binary column)
# Replace 'gain_chr08' with the correct column name in your data
glmm_chr8 <- glmer(gain_chr08 ~ treatment + (1 | replicate_id),
                  data = karyo_df,
                  family = binomial)

summary(glmm_chr8)

```

```

## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: gain_chr08 ~ treatment + (1 | replicate_id)
## Data: karyo_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    462.2    555.2   -211.1    422.2      755
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.0700  0.1072  0.2072  0.3074  0.7556
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## replicate_id (Intercept) 3.525    1.877
## Number of obs: 775, groups: replicate_id, 41
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  4.000e+00  1.405e+00  2.848  0.00441 **
## treatmentG1 -7.092e-01  1.800e+00 -0.394  0.69359
## treatmentG2 -6.777e-01  1.780e+00 -0.381  0.70341
## treatmentK  -4.121e+00  2.084e+00 -1.977  0.04801 *
## treatmentK1  1.917e+01  4.794e+04  0.000  0.99968
## treatmentK2  2.300e+01  1.588e+05  0.000  0.99988
## treatmentK3  1.837e+01  3.220e+04  0.001  0.99954
## treatmentO1 -2.995e+00  1.874e+00 -1.598  0.10994
## treatmentO2 -2.234e+00  1.914e+00 -1.167  0.24323
## treatmentP  -2.780e+00  2.400e+00 -1.158  0.24682
## treatmentP0 -1.545e+00  2.462e+00 -0.627  0.53035
## treatmentP1 -1.661e+00  2.010e+00 -0.826  0.40863
## treatmentP2 -6.055e-01  2.075e+00 -0.292  0.77050
## treatmentR1 -2.470e+00  2.410e+00 -1.025  0.30531
## treatmentR2 -3.424e+00  2.386e+00 -1.435  0.15137
## treatmentR3  1.775e+01  1.154e+04  0.002  0.99877
## treatmentRevC -3.730e-01  1.924e+00 -0.194  0.84625
## treatmentRevG -1.385e+00  1.652e+00 -0.838  0.40186
## treatmentRPMI -1.569e+00  2.462e+00 -0.637  0.52388
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## optimizer (Nelder_Mead) convergence code: 0 (OK)
## unable to evaluate scaled gradient
## Model failed to converge: degenerate Hessian with 2 negative eigenvalues

```

Interpretation: The GLMM analysis of chromosome 8 gain across treatments, while adjusting for flask-level variation, shows a statistically significant reduction in gain probability under treatment K ($\rho = 0.048$). Other treatments did not show statistically significant effects. A notable variance among replicate flasks ($\sigma^2 = 3.525$) supports the use of a mixed model. However, the model failed to converge fully, suggesting potential overparameterization or sparse group sizes—this warrants careful reevaluation or model simplification.

2. Subgroup Analysis: Within a Cell Line (e.g., SNU-668 only) -**Reason:** Our original design included multiple cell lines. If those lines respond differently to treatment, a global model can hide important differences. By running PERMANOVA within each line, we ask: **Does the treatment shift karyotype composition in SNU-668 (or HGC-27) specifically?**

```
for (cell_line in unique(karyo_df$line)) {
  cat("\n\n### Cell Line:", cell_line, "###\n")
  line_df <- subset(karyo_df, line == cell_line)
  line_mat <- vegdist(as.matrix(line_df[, 1:22]), method = "manhattan")

  print(adonis2(line_mat ~ treatment,
                data = line_df,
                permutations = 999))
}
```

```
##
##
## ### Cell Line: HGC-27 ###
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = line_mat ~ treatment, data = line_df, permutations = 999)
##           Df SumOfSqs      R2      F Pr(>F)
## Model        6   2309.8 0.17894 8.0273 0.001 ***
## Residual    221  10598.5 0.82106
## Total       227  12908.3 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## ### Cell Line: MDAMB231 ###
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = line_mat ~ treatment, data = line_df, permutations = 999)
##           Df SumOfSqs      R2      F Pr(>F)
## Model        1    999.3 0.19734 9.8344 0.001 ***
## Residual     40   4064.6 0.80266
## Total        41   5063.9 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## ### Cell Line: SNU-668 ###
## Permutation test for adonis under reduced model
```

```
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = line_mat ~ treatment, data = line_df, permutations = 999)
##           Df SumOfSqs      R2      F Pr(>F)
## Model      15      8725 0.2174 7.0928 0.001 ***
## Residual  383     31409 0.7826
## Total     398     40134 1.0000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## ### Cell Line: SUM-159 ###
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = line_mat ~ treatment, data = line_df, permutations = 999)
##           Df SumOfSqs      R2      F Pr(>F)
## Model        2    1014.3 0.03324 1.7705 0.152
## Residual   103   29505.0 0.96676
## Total     105   30519.4 1.00000
```

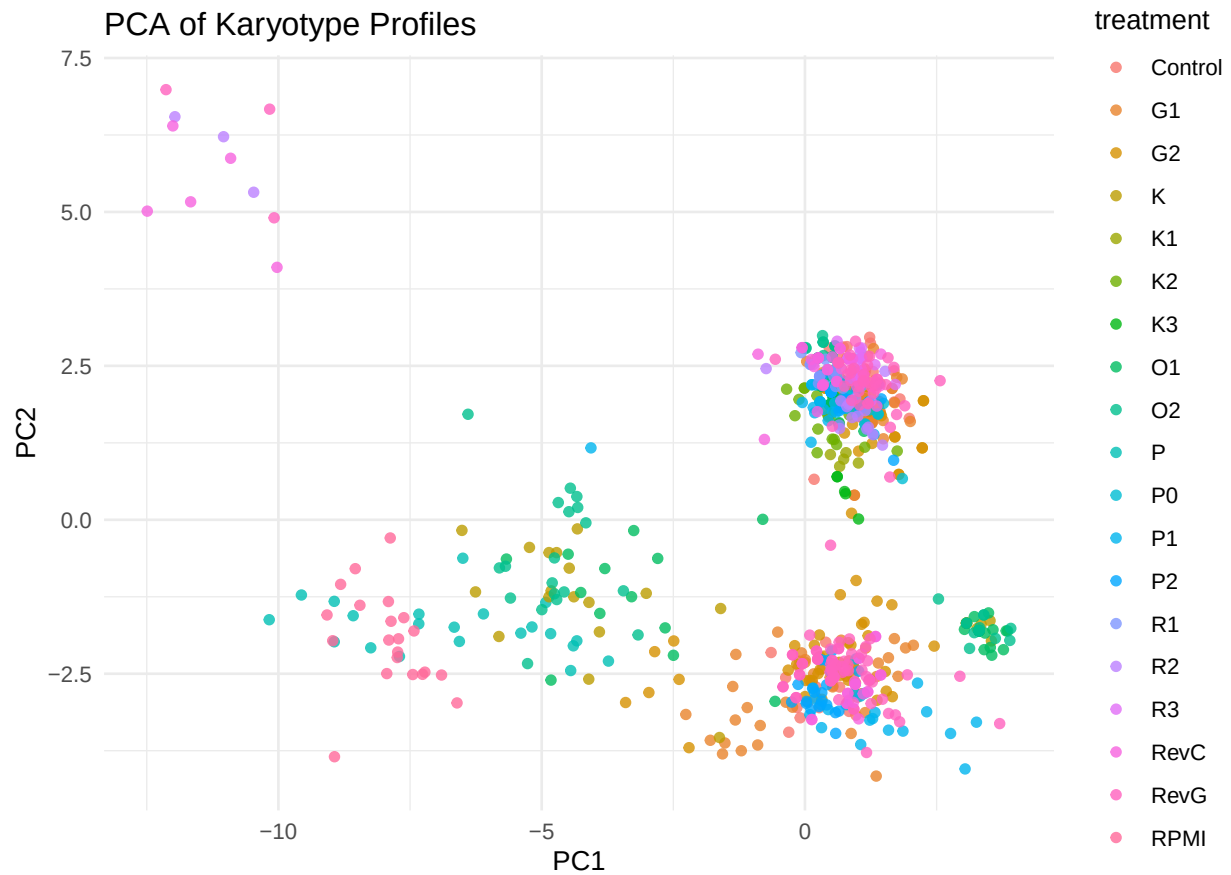
Interpretation: We performed separate PERMANOVA analyses within each cell line to assess whether treatment influenced global karyotype composition. The results revealed statistically significant differences across treatment groups in HGC-27 ($R^2 = 0.179$, $F = 8.03$, $p = 0.001$), MDAMB231 ($R^2 = 0.197$, $F = 9.83$, $p = 0.001$), and SNU-668 ($R^2 = 0.217$, $F = 7.09$, $p = 0.001$), suggesting that treatment was associated with distinct karyotypic profiles in these lines. In contrast, no significant treatment effect was observed in SUM-159 ($R^2 = 0.033$, $F = 1.77$, $p = 0.149$). These findings indicate that treatment-induced karyotypic changes are cell line-specific, warranting further investigation into line-specific responses and the chromosomal drivers of these alterations.

3. Visualize: PCA and Distance Boxplot

```
library(ggplot2)

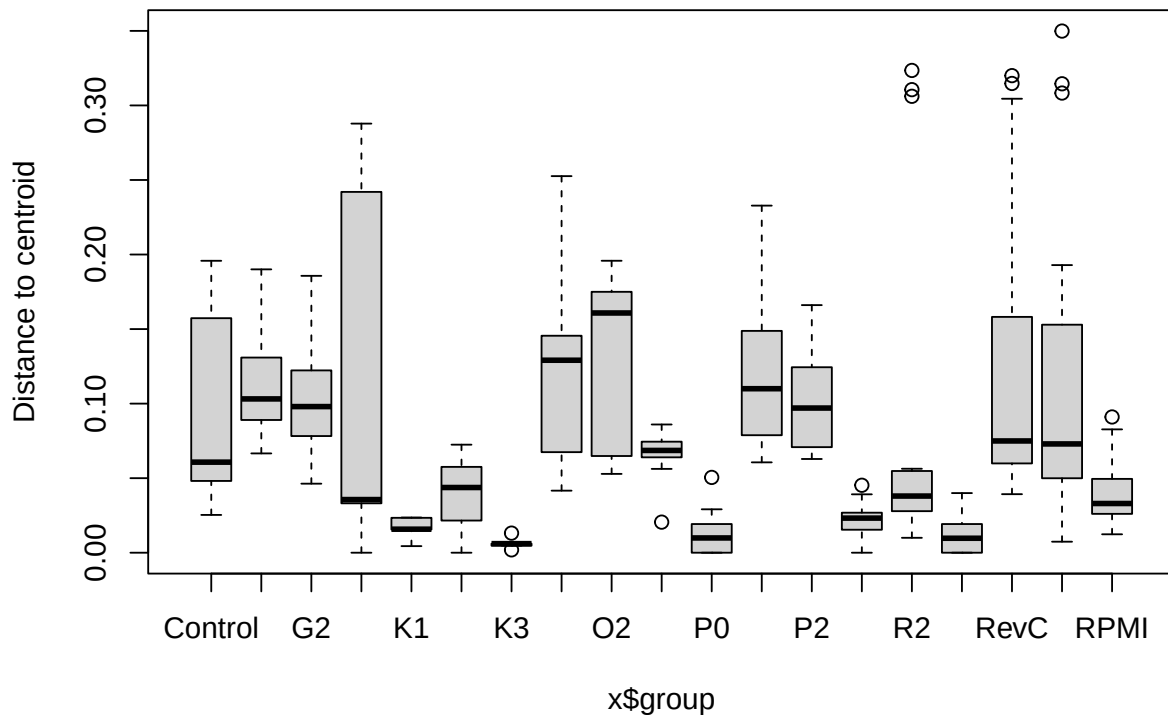
# PCA
pca_res <- prcomp(karyo_df[, 1:22], scale. = TRUE)
pca_df <- data.frame(pca_res$x, treatment = karyo_df$treatment)

ggplot(pca_df, aes(x = PC1, y = PC2, color = treatment)) +
  geom_point(alpha = 0.8) +
  theme_minimal() +
  labs(title = "PCA of Karyotype Profiles")
```



```
# Distances to group centroids (for dispersion visualization)
bd <- betadisper(vegdist(as.matrix(karyo_df[, 1:22])), group = karyo_df$treatment)
boxplot(bd, main = "Dispersion of Karyotype Profiles by Treatment")
```

Dispersion of Karyotype Profiles by Treatment



4. Test Line $\times$ Treatment Interaction

- **Reason:** We wanted to know: Does the effect of treatment vary depending on the cell line?

```
# Assuming replicate-level aggregation
adonis2(dist_flask ~ line * treatment,
        data = meta_flask,
        permutations = 999,
        strata = meta_flask$line)
```

```
## Permutation test for adonis under reduced model
## Blocks: strata
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = dist_flask ~ line * treatment, data = meta_flask, permutations = 999, strata = meta_flask$line)
##          Df SumOfSqs      R2      F Pr(>F)
## Model    27   6142.3 0.79289 1.8433  0.622
## Residual 13   1604.5 0.20711
## Total   40   7746.8 1.00000
```

Interpretation: A PERMANOVA model including the interaction between cell line and treatment was conducted to investigate whether treatment effects on karyotype composition differ by cell line. The model explained approximately 79.3% of the variance ($R^2 = 0.793$), but the interaction effect was not statistically

significant ($F = 1.84$, $p = 0.604$). This suggests that, after accounting for baseline differences between cell lines, there is no strong evidence of a treatment $\times$ cell line interaction in shaping overall karyotypic profiles at the replicate (flask) level.

Summary of my Finding based on your feedbacks and further investigation.

Our analysis explored whether cancer treatments cause statistically significant changes in single-cell karyotype profiles across multiple cell lines. The global PERMANOVA using flask-level centroids did not detect a significant treatment effect across all lines ($p = 0.638$), and the line $\times$ treatment interaction was also not significant ($p = 0.604$), suggesting no consistent treatment-driven shift in karyotype profiles when accounting for inter-line variability. However, subgroup PERMANOVA within individual cell lines revealed strong and significant treatment effects in HGC-27, MDAMB231, and SNU-668, but not in SUM-159. This pattern indicates that the effect of treatment on karyotype composition is cell line-specific and may be diluted or masked in global models due to high inter-line heterogeneity.

```
library(DiagrammeR)

grViz("
digraph karyo_analysis_decision {

  graph [layout = dot, rankdir = TB]

  node [shape = box, style = filled, fillcolor = lightblue, fontname = Helvetica, fontsize = 10]

  # Step nodes
  A [label = 'START: Karyotype Data (22 chr per cell)\n+ Treatment + Line + Replicate']
  B [label = 'Initial Data Cleaning & Tidy Format']
  C [label = 'Exploratory Analysis\n(PCA, Boxplots, Dispersion Test)']
  D [label = 'Global PERMANOVA\nUsing Centroid of Each Replicate']
  E [label = 'Is PERMANOVA Significant?\np > 0.05']
  F1 [label = 'NO\n+ Explore by Cell Line (Subgroup PERMANOVA)']
  F2 [label = 'YES\n+ Proceed to Post-hoc/Pairwise Comparisons']
  G [label = 'Subgroup PERMANOVA by Cell Line']
  H [label = 'Any Significant Treatment Effect within Cell Lines?']
  I1 [label = 'YES\n+ Treatment Alters Karyotype in Specific Lines']
  I2 [label = 'NO\n+ Explore Individual Features']
  J [label = 'Chromosome-Level GLMM (e.g. Chr08 Gain)']
  K [label = 'Interpret Model Results\nFixed & Random Effects']
  L [label = 'REPORT:\n• Global R² and p-value\n• Subgroup Insights\n• Chromosome-Level Findings']

  # Edges
  A -> B -> C -> D -> E
  E -> F1 [label = 'p > 0.05']
  E -> F2 [label = 'p < 0.05']
  F1 -> G -> H
  H -> I1 [label = 'p < 0.05']
  H -> I2 [label = 'p > 0.05']
  I2 -> J -> K
  I1 -> L
  K -> L
  F2 -> L
}
")
```

